FleetRoute: Bounded-Authority Route Optimization for Auditable Real-Road Fleet Operations

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Abstract

Operational route optimization can fail even when the returned route is feasible and low-cost: the route may be correct for a planning state that is no longer authorized for execution. FleetRoute makes *state identity* and *publication authority* part of routing correctness. At solve invocation, it freezes a checksum-identified OpenStreetMap-derived directed graph together with the depot, work points, fleet and timing rules, objective, and road revision. Six stochastic software workers may propose private candidates, but none can publish. A candidate becomes public only after independent replay reconstructs feasibility and objective value on the frozen graph and a protected commit confirms that the contract remains current. Driver-facing surfaces consume committed revisions only; an exact-edge incident remains pending until operator approval advances the road revision and triggers replanning.

Evaluation tests three distinct claims: optimizer competence, end-to-end state coherence, and packaged-release reliability. The frozen campaign contains 198 configured runs over 22 instances: 18 public CVRP/VRPTW benchmarks and four immutable directed-road graph/input snapshots derived from regional OpenStreetMap extracts in Thailand. The final frozen solver generation is feasible on 66/66 configured runs, is best or tied-best in 47/51 strict complete instance/seed triples, and has a 2.02% mean best-of-three gap to public references. On the four frozen road graphs, its three-seed mean matches the normalized Google OR-Tools best-of-12 baseline at 50 service nodes and is lower by 1.10%, 2.73%, and 1.51% at 100, 300, and 450 nodes under the same independent evaluator. Controlled Thailand and Phuket workflows exercise the publication chain from planning through field feedback. FleetRoute remains pre-deployment: the contribution is an auditable execution-authority architecture with reproducible evidence lineage, not a claim of realized field impact.

Operational Problem: Correct Route, Wrong State

Vehicle-routing research typically asks whether a route is feasible and good for a specified instance (Dantzig and Ramser 1959; Solomon 1987). An operational system must first ask a different question: *is this still the instance authorized for execution?* A cached graph can belong to the wrong region; a transfer can be incomplete; road metadata can be missing; work points can change while search is running; and a driver can report a closure after publication. Dynamic-routing research recognizes that new information changes the decision problem (Pillac et al. 2013), while waste-routing studies emphasize that data quality and operating rules determine practical usability (Beliën, De Boeck, and Van Ackere 2014; Hess et al. 2024). FleetRoute therefore treats route quality and plan-state integrity as one operational correctness problem.

FleetRoute originated as a waste-collection optimizer and evolved into a single-depot decision-support application for homogeneous local fleets. Work-point quantity can represent collection load or depot-dispatched delivery load; service duration and optional time windows encode stop-level requirements. The scope deliberately excludes mixed pickup-and-delivery precedence, split delivery, live traffic, heterogeneous compartments, and vehicle-specific legal clearance. This restriction is intentional: it permits a stronger invariant in the implemented system—every public route is traceable to the exact road source, work-point set, fleet rules, objective, and incident revision against which it was independently certified.

FleetRoute does not introduce a new VRP formulation. Its contribution is an execution-authority architecture around optimization, built from four coupled elements: **source-bound planning state**, **bounded optimizer authority**, **human-governed road revision**, and **evidence lineage**. The governing rule is deliberately restrictive: *a solver result is a proposal, not a public plan*. Independent certification and protected commit decide what may become operational. A late, stale, or unverifiable stochastic result can therefore be rejected without silently replacing the trusted plan. The evaluation follows the same boundary by separating optimization competence, input-to-field state coherence, and release reliability; field impact is left to a pilot rather than inferred from offline evidence.

FleetRoute as a Bounded-Authority Decision System

Road Identity and a Frozen Planning Contract

Planning begins with a regional OpenStreetMap (OSM) extract acquired from Geofabrik (Geofabrik GmbH 2026; OpenStreetMap Contributors 2026). FleetRoute reads

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OSM's Protocolbuffer Binary Format (PBF) and materializes a directed road graph $G = (V, E)$ whose vertices are routable road-network nodes and whose directed edges encode permitted travel direction, length, and static metadata used for travel-time preparation. Direction is operationally significant: the cost, or even existence, of $u \to v$ need not match $v \to u$. The PBF checksum is verified and bound to dependent graph caches and run evidence. A region becomes active only after a coverage gate verifies the current depot and work points; selecting an incompatible source invalidates dependent state instead of silently reusing an old graph. Tagged `maxspeed` values take precedence, with explicit highway-class fallbacks when metadata are absent.

Map-created or CSV-imported work points carry role, name, coordinates, demand, service duration, and optional earliest/latest service times. At solve invocation, FleetRoute freezes the planning contract

$$C = (h_G, d, W, F, O, r),$$

where h_G identifies the road source and graph, d is the depot, W the work-point set, F the fleet/timing rules, O the objective, and r the road-revision state. Eligibility is contract-specific: a worker that finishes later acquires no authority over a newer state.

Hybrid Search Proposes; Independent Replay Certifies

The native C++ engine uses portfolio-style hybrid search: constructive starts, simulated annealing, adaptive and fragment large-neighborhood search, route-local and inter-route improvement, and elite/diversification mechanisms, consistent with established hybrid VRP practice (Ropke and Pisinger 2006; Vidal et al. 2014; Wouda, Lan, and Kool 2024). Six configured search instances act as independent stochastic software workers; this is a software-concurrency setting, not a claim of six physical CPU cores. Candidate lists, road-cost caching, top-k insertion, and dirty-route recomputation reduce repeated work.

The optimizer is deliberately non-authoritative. Heuristic search may maintain provisional states, but it cannot certify its own publication. For each candidate considered for commit, FleetRoute reconstructs the route in an objective-matched context and checks depot structure, customer coverage and uniqueness, capacity, directed reachability, service timing, time windows, distance, duration, and the claimed objective. On road instances, terminal-to-terminal costs are replayed from the serialized directed graph rather than copied from solver output. Publication therefore requires

$$\text{publish}(x) \Rightarrow \text{certify}(x, C) = \text{PASS} \ \wedge \ C = C_{\text{current}}.$$

Road-cost preparation, governed search, and final certification are timed separately so preprocessing or validation cannot be concealed inside a nominal search budget.

Protected Commit and Human-Governed Replanning

A solve attempt never writes directly to public route state. It runs in private attempt state, and cancellation, timeout, malformed output, validation failure, incomplete evidence, stale state, or an inferior candidate leaves the protected incumbent unchanged. A valid rerun can remain in history without displacing the public plan. This transactional boundary lets optimization fail without erasing the last trusted route revision.

Field changes cross the same authority boundary. A driver reports the affected road *segment*, not a generic map point; the report first enters a pending operator state. Approval advances the road revision, invalidates affected dependencies, and triggers a fresh solve-certify-commit cycle. Until that cycle succeeds, the incumbent remains authoritative. Driver manifests, playback, exports, and navigation handoff consume committed revisions only; external navigation can assist turn-by-turn travel but does not acquire authority to reorder FleetRoute's published stop sequence.

Engineering the Authority Boundary

Each major control in FleetRoute closes a failure mode exposed during construction. Missing usable `maxspeed` metadata broke graph preparation, motivating an explicit highway-class fallback hierarchy. Stateful interaction exposed a more consequential hazard: a worker can return a perfectly feasible route for an obsolete region, depot, stop set, objective, or incident revision. That led to solve-time contract freezing and a freshness test at commit. Directly overwriting route output made cancelled or inferior reruns destructive, leading to private attempt state and atomic promotion. Later hardening exposed failures outside the optimizer—interrupted PBF transfers and concurrent build/install work—which motivated resumable transfer, checksum revalidation, and shared locks. The recurring pattern is architectural: once optimization is embedded in an interactive application, correctness depends on the transitions around search as much as on search quality. FleetRoute makes those transitions explicit as identities, contracts, certification gates, revisions, and receipts.

Evidence: Optimization, State Coherence, and Release Reliability

The evaluation mirrors the system's authority boundaries and keeps three claims separate. *Optimization competence* asks whether search quality is adequate for the application. *State coherence* asks whether one road/workload revision remains intact from input through certification, publication, and field consumption. *Release reliability* asks whether the packaged application preserves those controls under regression fixtures. These questions require different evidence, and none is treated as a proxy for realized organizational impact. The appendices identify the evidence authority behind every quantitative statement.

Frozen Benchmark and Cross-Generation Evidence

The frozen solver campaign contains two complementary evidence sets. **Public benchmarks** comprise 18 standard instances: three Augerat-A CVRP cases, six Solomon VRPTW cases, six 200-customer Gehring–Homberger VRPTW cases, and three CVRPLIB-X CVRP cases. Each family is

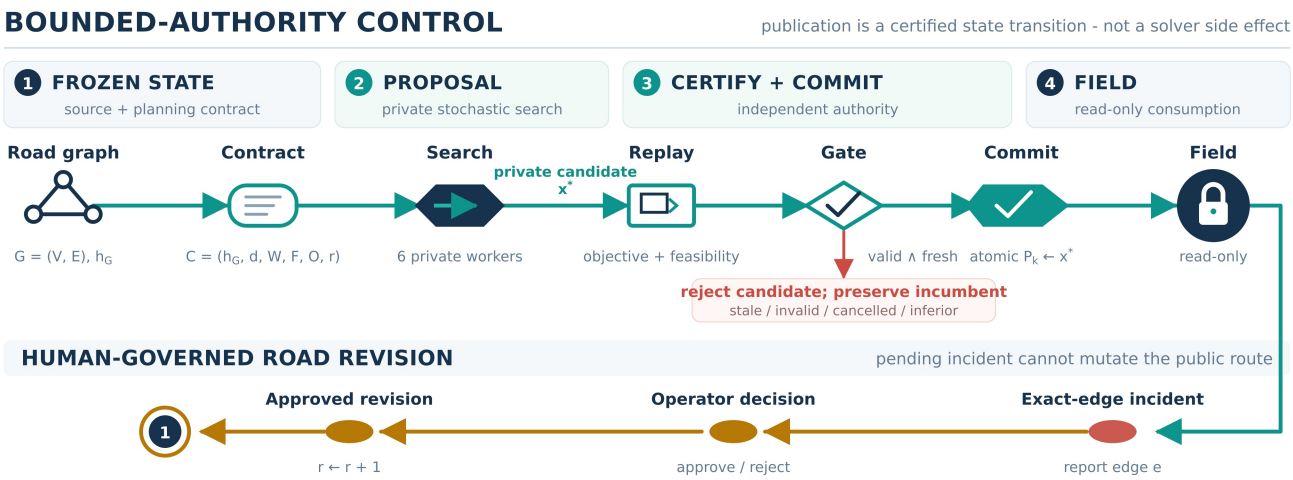


Figure 1: Bounded-authority architecture. The upper band separates frozen decision state, private search, independent replay, certification, protected publication, and read-only field use. Search can only propose a private candidate; publication requires independent feasibility replay and contract freshness, while rejection preserves the incumbent. The lower band shows an exact-edge incident entering planning state only after operator approval advances the road revision and forces a new solve-certify-commit cycle.

Operational pressure	Tempting shortcut	Failure exposed	Control adopted / transferable lesson
Road acquisition	Trust the latest cache or region label.	A valid graph can represent the wrong geography; partial transfer or incomplete metadata can break preparation.	Verify source identity and coverage; resume and recheck transfers; make metadata fallbacks explicit.
Concurrent edits	Accept any feasible worker result.	Feasibility can hold for an obsolete depot, stop set, objective, or road revision.	Freeze the request contract and reject stale candidates at commit.
Rerun/cancellation	Overwrite the public route.	A failed, cancelled, or inferior attempt can destroy the only trusted plan.	Run privately; certify first; atomically promote or preserve the incumbent.
Search telemetry	Treat worker-local improvement as public truth.	Progress can overstate what has actually been validated.	Expose validated global-best evidence and separate preparation, search, and certification timing.
Driver road change	Report a point or defer plan authority to navigation.	The affected edge is ambiguous and route order can drift outside the audited plan.	Select an exact segment, require operator approval, and keep navigation downstream.

Table 1: Controls derived from observed failure modes. Across acquisition, search, publication, telemetry, and field feedback, the common design rule is to make mutable state explicit before optimization is allowed to alter a public plan.

evaluated against its archived instance and frozen reference objective. For reference objective z^{ref} and FleetRoute's best objective over three configured seeds z^{best}, the reported gap is $100(z^{\mathrm{best}} - z^{\mathrm{ref}})/z^{\mathrm{ref}}$.

Real-road benchmarks are four frozen optimization inputs derived from regional OpenStreetMap extracts in Thailand. Each case serializes an immutable directed graph with depot and service-node data; no live map query participates in evaluation. Legacy, V3.1, Current, and the OR-Tools bridge therefore see the same road nodes, directed edges, edge lengths, depot, and non-depot service nodes. The four cases contain 50, 100, 300, and 450 service nodes. Tables retain the conventional term "customer," but here it denotes a non-depot optimization node, not a real customer organization. Route quality is static directed-road distance reconstructed from the frozen graph rather than straight-line distance between coordinates.

Legacy, **V3.1**, and **Current** are successive archived generations of the same FleetRoute solver family, not external algorithms. Legacy is the earliest preserved baseline, V3.1 the intermediate generation, and Current the final generation frozen for this campaign. All use seeds 777, 1777, and 2777. Later v8.1 application hardening is evaluated separately and cannot retroactively change the frozen benchmark. Because each instance has only three configured seeds, cross-generation statistics are descriptive rather than inferential.

Compute allowance is fixed *ex ante*. For each instance, Legacy, V3.1, and Current receive the same three seeds, six configured software search workers (`-instances 6`), and the same hard algorithmic ceiling. All 198 archived parent commands set `-no_improve_stop_s 0`, so the scored comparison has no no-improvement patience cutoff. Per-seed ceilings are 60 s for Augerat-A, 90 s for Solomon, 120 s for Gehring–Homberger-200, 90/120/180 s for the three CVRPLIB-X sizes, and 60/120/170/350 s for the 50/100/300/450-node road cases. A process can finish early, and post-deadline evidence emission is recorded separately; the configured ceiling, not observed completion time, defines the cross-generation search allowance.

Current is valid on all 66 configured runs (22 instances $\times$ 3 seeds). Across the 18 public instances, its best-of-three gap to the frozen reference averages 2.02%; 13/18 are within 3%, 17/18 within 5%, and the largest gap is 6.07%. On CVRPLIB-X, Current records 27,715 versus 27,591 on X-n101-k25 (0.45%), 60,198 versus 58,578 on X-n200-k36 (2.77%), and 22,581 versus 21,736 on X-n303-k21 (3.89%).

Strict cross-generation claims use only the 51 parent-original instance/seed triples in which all three generations are valid. Current is best or tied-best in 47/51 (92.2%) and has 0.19% mean excess above the best generation within each triple, compared with 2.22% for V3.1 and 11.23% for Legacy. Pairwise, Current is better/tied/worse than V3.1 in 38/9/4 triples and than Legacy in 42/9/0. A separate one-shot recovery protocol yields 17 valid routes from 18 attempts, raising descriptive coverage from 180/198 to 197/198. Recovery wall time is excluded from timing claims, and Legacy X-n303-k21 seed 777 remains unresolved rather than imputed.

Google OR-Tools (Google 2026) provides an external solution-quality baseline on the four directed-road cases, not a runtime comparator. Twelve routing configurations are run per case. "Normalized OR-Tools best-of-12" is the lowest objective after selected route sequences are replayed by the same independent directed-road evaluator used for FleetRoute; raw OR-Tools internal costs are never mixed with FleetRoute scores. The historical OR-Tools campaign used per-configuration limits of 30/175/400/1000 s at 50/100/300/450 nodes, giving 12-configuration allowance ceilings of 360/2100/4800/12000 s. Retained raw output for the 100-node sweep records all 12 variants and 1214.022 s total elapsed; complete sweep elapsed logs are not retained for the other cases, so only their configured allowances are reported. Current's three-seed mean matches the normalized baseline at 50 nodes and is lower by 1.10%, 2.73%, and 1.51% at 100, 300, and 450 nodes. Project records place both campaigns in Google Colab on Intel Xeon 2.4 GHz-class hosts; this contextualizes compute but does not normalize it. Because search structure and stopping behavior differ, the claim is strictly common-metric solution quality, not runtime superiority.

Controlled End-to-End Workflows

The larger controlled workflow uses a verified regional Thailand road source with the depot at Prince of Songkla University (PSU) in southern Thailand. Two hundred synthetic work points are sampled within approximately 7 km of the depot and validated for road access on the selected directed graph before optimization. The scenario is a controlled workflow stress test at realistic map scale, not a municipal field dataset. FleetRoute publishes 22 routes, and the same committed revision is consumed by validated convergence evidence, the fleet map, route cards, driver manifests, schedule-aware playback, and run history. The captured job records 2.6 s of road-cost preparation and 45.2 s of governed search within 48.5 s elapsed. The result demonstrates state coherence at nontrivial planning scale: source identity, graph preparation, search, certification, publication, audit evidence, and driver-facing consumption remain bound to one public revision. It does not measure realized field savings.

A second controlled scenario in Phuket begins with a four-row CSV: one depot and three work points with role, name, latitude, longitude, demand, service duration, and earliest/latest service times. The rush workflow validates the rows, binds them to accessible directed-road locations, prepares road costs, and publishes an initial route. A driver then reports an exact road segment. The incident becomes a pending operator decision; approval creates a new road revision and triggers a fresh solve, while the incumbent remains authoritative until a replacement is independently certified and committed. Only then can the driver surface hand the committed stop order to Google Maps. The scenario isolates the full authority chain—structured input, planning, field change, human decision, replanning, certification, and downstream navigation—without treating developer-operated interaction as real-user deployment.

Application-Hardening Regression

Application hardening is evaluated separately from the frozen solver campaign so a later software snapshot cannot leak into algorithmic evidence. The v8.1 regression audit covers the 48-file embedded application manifest; Python, JavaScript, and C++17 syntax; exact road-access acceleration against exhaustive queries; 450-stop bulk import; graph materialization and reuse; interrupted PBF-transfer resumption; shared build/install locks; cold API startup; and a controlled zero-to-published-route workflow. In the 450-customer fixture, 451 road-access validations take 0.426 s, first materialization 0.250 s, and cache reuse 0.017 s. Across 100 exactness comparisons, the maximum accelerated-versus-exhaustive road-access distance difference is 1.16×10^{-10} m. A controlled 40-customer zero-to-route test publishes route version 2 with four independently validated routes. These measurements support release reliability, not generalized production throughput.

Deployment Boundary and Transferable Lessons

The applied question is broader than "shorter routes." For every public plan, FleetRoute is designed to answer three operational questions: *what exact state was optimized, what authorized the transition to this revision, and what evidence certifies the result?* A late stochastic result can be rejected without destroying the trusted incumbent; a driver-reported

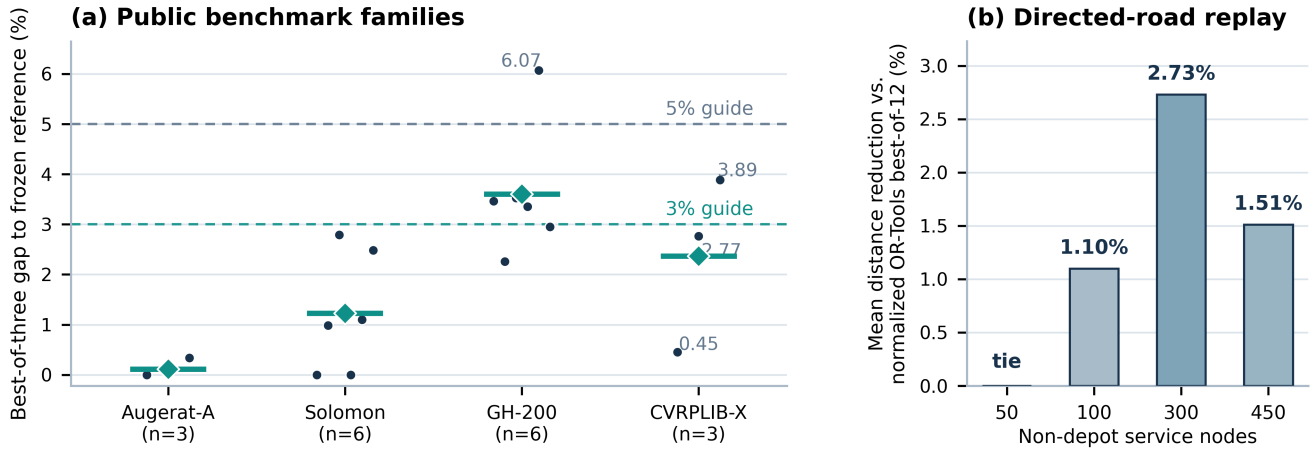


Figure 2: Solution quality under frozen evaluation contracts. (a) Current's best-of-three gap to the frozen reference across 18 public CVRP/VRPTW instances; points are instances and diamonds/segments summarize family means. (b) On four frozen directed OpenStreetMap-derived road graphs from Thailand, Current's three-seed mean is compared with the best of 12 OR-Tools route sequences after independent common-evaluator replay. The comparison is route quality under a common metric; native solver cost and runtime are not normalized.

road change can affect routing only through an explicit operator decision. These are failure modes that objective-value benchmarks do not measure, yet they become central as soon as optimization is connected to field use.

FleetRoute remains an Emerging Application rather than a production deployment. Persistent organizational use would require managed hosting, authentication and role separation, durable storage, monitoring, backup, operational ownership, and local validation of vehicle restrictions. OpenStreetMap can be incomplete or stale and does not certify legal or physical access for a particular vehicle (OpenStreetMap Contributors 2026; Boeing 2017); static speeds are not live traffic. The demonstrations are developer-operated, use synthetic work points in the 200-stop Thailand case, and do not measure realized operating cost, driver adoption, or longitudinal failure rates.

The next evidence step is a bounded service-region pilot with an organization that already plans recurring routes. It should compare planner preparation time, vehicles used, realized distance/time, late or missed stops, and operating cost with the incumbent process; record stale or invalid candidates blocked before publication; measure incident-to-certified-plan latency and failed replans; and capture route-order deviation, navigation error, task completion, training burden, and usability feedback. A conservative progression is appropriate: validate one recurring region under human dispatch supervision, then add managed hosting, authentication/role separation, durable storage, monitoring, backup, and local vehicle-access rules before broader use. The evaluation target is explicitly two-dimensional: optimization benefit *and* decision-pipeline reliability.

Three lessons extend beyond FleetRoute. **First, data identity is part of decision state**: a route cannot be trusted independently of the source and revision that produced it. **Second, optimization should have bounded authority**: search may propose, but independent certification and transactional commit should decide what becomes public. **Third, evidence lineage should survive application hardening**: interface, cache, and orchestration improvements must not silently rewrite the benchmark cited by the paper. The frozen benchmark establishes sufficient optimizer competence for the application; the stronger applied contribution is the authority structure around search. Field impact is the next evidence step, not a conclusion of the present study.

Reproducibility and Evidence Lineage

Every quantitative claim has a designated evidence authority. Public-benchmark quality claims use frozen instance identities and family-specific reference objectives. Real-road comparisons use SHA-256-identified serialized OSM-derived graph inputs; complete graph/instance snapshots are stored and reused unchanged rather than rebuilt from a live service, and a common evaluator replays every compared route sequence on those exact directed graphs. Workflow claims use captured committed-state traces; hardening claims use regression fixtures applied to the packaged application.

The Code & Data Supplement includes the exact runnable v8.1 Google Colab notebook and its decoded source snapshot. A reviewer can execute the notebook's two code cells in order: the application payload is embedded, release identity is verified before launch, and no repository checkout is required. The supplied verifier byte-compares the notebook payload, source snapshot, release manifest, and native solver source. This separation is methodological, not cosmetic: benchmark quality is not evidence of operational readiness, and interface polish is not evidence of solver qual-

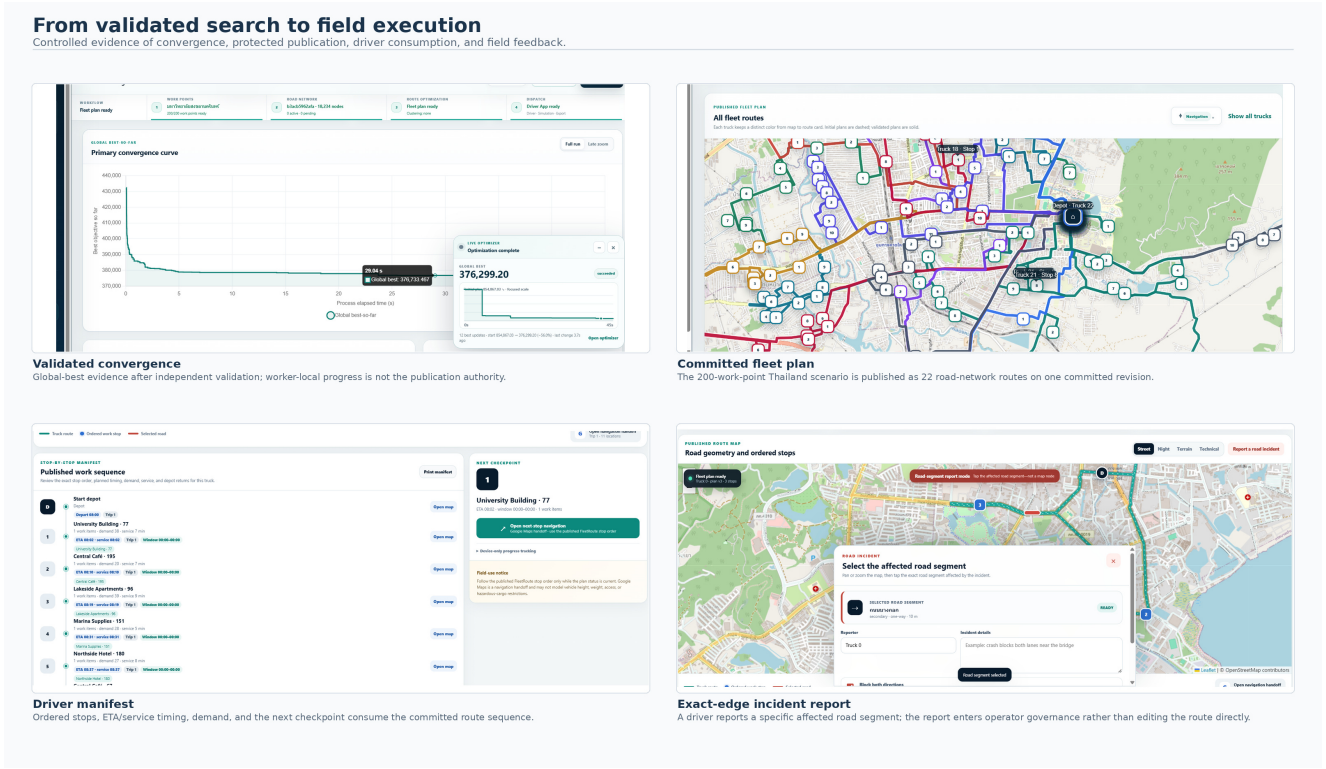


Figure 3: Controlled evidence from validated search to field feedback: validated global-best convergence, the committed 22-route Thailand plan, a driver manifest consuming the committed stop order, and exact-road-edge incident reporting. The panels show optimization evidence, publication, driver-facing consumption, and field feedback as distinct states of one governed route revision rather than independent UI views.

ity. FleetRoute therefore preserves frozen artifacts, names the authority behind each quantitative claim, and keeps residual uncertainty explicit. Realized organizational benefit, driver adoption, and longitudinal reliability require a field pilot.

Appendix A: Evidence Boundary, Lineage, and Verification

The appendices provide a reviewer-facing inspection trail: evidence boundaries, benchmark-detail tables, replay summaries, reliability regressions, and operational screenshots. They are descriptive and audit-oriented; the main paper defines the claim scope, while the appendices make the supporting lineage easier to inspect and reproduce.

The application artifact is FleetRoute v8.1 SYSTEM_HARDENED. Its native optimizer source is byte-identical to the prior application solver snapshot in the frozen experimental lineage. The Code & Data Supplement contains the exact runnable notebook `app/FleetRoute_International_MapFirst_FINAL_WORKFLOW_AUDITED_v8_1_SYSTEM_HARDENED.ipynb`, its decoded source snapshot, immutable parent and targeted-recovery benchmark archives, the common evaluator, OR-Tools bridge, frozen OSM input identities, rebuilt tables, manifests, and verification utilities. The notebook contains one explanatory Markdown cell and two executable code cells; running the two code cells in order in Google Colab reproduces the submitted application workspace, subject to normal network availability for missing prerequisites, public road-data acquisition, and the temporary tunnel.

The release receipt records 213/213 tracked supplement files matching the root SHA-256 manifest. Application verification checks the exact submitted notebook, its embedded 48-file release manifest, decoded source snapshot, and native solver hash. Parent verification checks the 198-run/22-instance identities and benchmark solver hashes. Independent common-evaluator replay reproduces objective, route count, lateness, and validity for all 180 originally valid parent routes. Strict paper claims remain parent-original. Targeted recovery contributes 17 valid routes from 18 attempts, yielding 197/198 descriptive solution-quality coverage; recovery wall time is excluded from timing comparisons.

Evidence layer	Frozen unit, purpose, and claim boundary
Parent benchmark	198 configured runs / 22 instances; authority for strict cross-generation and timing claims; parent-original observations only.
Targeted recovery	18 one-shot attempts; restores descriptive solution-quality coverage; recovery timing is excluded.
Application release	v8.1 notebook + 48-file payload; supports workflow and hardening claims; does not retroactively change the frozen solver benchmark.
Independent replay	180/180 parent-valid routes; verifies common objective and feasibility; re-scores sequences but creates no new search result.
Controlled UI evidence	Thailand and Phuket workflows; demonstrates state transitions and user surfaces; not a production-impact or real-user study.

Table 2: Evidence hierarchy and claim boundary.

Appendix B: Public Benchmark and CVRPLIB-X Detail

The benchmark contract contains 22 instances. Four are FleetRoute real-road cases built from regional OpenStreetMap (OSM) Protocolbuffer Binary Format (PBF) extracts of Thai provinces and frozen as immutable directed graph/input snapshots with 50, 100, 300, and 450 non-depot service nodes. Because an OSM-derived road segment can be absent or have a different cost in the reverse direction, the benchmark objective is static directed-road distance on the frozen graph rather than straight-line distance. The remaining 18 instances are public benchmarks: three Augerat-A CVRP, six Solomon VRPTW, six Gehring–Homberger 200-customer VRPTW, and three CVRPLIB-X CVRP cases. Every solver generation uses seeds 777, 1777, and 2777. Legacy, V3.1, and Current identify successive archived versions of the same solver family rather than external baselines. The independent common evaluator, not a native solver log, is authoritative for cross-generation objective comparison.

CVRPLIB-X is also reported separately so performance on the larger selected public cases is visible rather than absorbed into a global average.

Instance	Reference	Best	Mean	Best gap
Augerat-A				
A-n32-k5	784.0	784.0	784.0	0.00%
A-n60-k9	1354.0	1354.0	1354.7	0.00%
A-n80-k10	1763.0	1769.0	1778.7	0.34%
Solomon				
C101	827.3	827.3	827.3	0.00%
R101	1637.7	1653.8	1658.9	0.98%
RC101	1619.8	1665.0	1669.1	2.79%
C201	589.1	589.1	589.1	0.00%
R201	1143.2	1155.8	1163.3	1.10%
RC201	1261.8	1293.1	1303.8	2.48%
Gehring–Homberger 200				
C1_2_1	2698.6	2792.0	2797.7	3.46%
C2_2_1	1922.1	1965.5	1980.9	2.26%
R1_2_1	4667.2	4831.9	4880.1	3.53%
R2_2_1	3468.0	3584.3	3632.3	3.35%
RC1_2_1	3516.9	3730.4	3758.5	6.07%
RC2_2_1	2797.4	2879.9	2923.1	2.95%
CVRPLIB-X				
X-n101-k25	27591	27715	27863.0	0.45%
X-n200-k36	58578	60198	60363.0	2.77%
X-n303-k21	21736	22581	22621.3	3.89%

Table 3: Current-solver public benchmark detail. Best and mean are computed over the three configured seeds under each family's frozen metric.

Family	n	Cap/seed	Frozen scoring
Augerat-A	3	60 s	CVRP EUC_2D
Solomon	6	90 s	DIMACS 0.1
GH-200	6	120 s	DIMACS 0.1
CVRPLIB-X	3	90/120/180 s	CVRP EUC_2D

Table 4: Public-suite compute and scoring contract. CVRP distance uses nearest-integer EUC_2D; Solomon and Gehring–Homberger use the 0.1-truncated DIMACS convention for distance and travel time. Every row uses seeds 777/1777/2777, six configured software workers, and no no-improvement early stop.

Instance	Solver	Valid	Best	Mean	Best gap
X-n101-k25	Current	3/3	27715	27863.0	0.45%
	V3.1	3/3	28129	28260.7	1.95%
	Legacy	3/3	28196	28323.0	2.19%
X-n200-k36	Current	3/3	60198	60363.0	2.77%
	V3.1	3/3	60237	60316.0	2.83%
	Legacy	3/3	61509	61933.0	5.00%
X-n303-k21	Current	3/3	22581	22621.3	3.89%
	V3.1	3/3	22590	22714.7	3.93%
	Legacy	2/3	28187	28562.5	29.68%

Table 5: CVRPLIB-X cross-generation detail. Legacy X-n303-k21 seed 777 is the sole unresolved run after targeted recovery and is not imputed.

Appendix C: Real-Road Replay and Cross-Generation Validity

Execution environment. The frozen parent receipt records Linux x86_64, two logical CPUs visible to the benchmark process, g++ 11.4.0, C++17, and `-O3`. The `-instances 6` setting therefore denotes six software search workers multiplexed over the runtime-visible CPUs, not six physical cores. Contemporaneous project documentation describes the FleetRoute and OR-Tools campaigns under a Google Colab / Intel Xeon 2.4 GHz-class protocol. This context is disclosed to bound compute interpretation; no hardware-independent speedup is claimed.

The four OpenStreetMap-derived inputs originate from regional Thailand PBF extracts and carry SHA-256 identities in the supplement. Legacy, V3.1, Current, and the OR-Tools bridge are therefore evaluated on the same immutable directed-road instances. "OR-Tools" in Table 6 denotes the best of 12 route sequences after common-evaluator replay, not a raw OR-Tools internal objective. Fleet caps are 10, 20, 60, and 100; capacity is 200 for all four cases and the route clock begins at 08:00. Table 7 discloses compute allowance without turning the comparison into a speed claim.

Overall feasibility in the frozen/recovered evidence is 66/66 for Current, 66/66 for V3.1, and 65/66 for Legacy. The strict parent-original analysis contains 51 complete triples. Recovery-augmented descriptive analysis contains 65 complete triples, 14 of which include at least one recovered observation. The sole unresolved row is CVRPLIB-X / X-n303-k21 / Legacy / seed 777; it ended in an external timeout and remains explicitly absent.

The strict parent-original analysis contains 51 instance/seed triples with valid results from all three generations. Pairwise comparison makes the aggregate best/tied count explicit: Current is better/tied/worse than V3.1 in 38/9/4 triples and better/tied/worse than Legacy in 42/9/0. Averaged over these complete triples, Current reduces distance by 1.90% relative to V3.1 and 8.41% relative to Legacy when each triple is normalized by the comparator's distance. These are paired descriptive statistics over the frozen selected suite, not a claim of universal dominance.

N	OR-Tools	Current best	Current mean	Mean gain
50	32737.521	32737.521	32737.521	0.00%
100	123102.066	121637.565	121748.416	1.10%
300	336733.176	325360.576	327535.179	2.73%
450	778051.453	763297.125	766294.016	1.51%

Table 6: OpenStreetMap-derived directed-road route quality (meters) under the independent common evaluator. OR-Tools denotes the normalized best-of-12 baseline; differing search protocols preclude a runtime comparison.

N	FL cap/seed	OR cap/config	OR sweep cap	OR elapsed
50	60	30	360	–
100	120	175	2100	1214.022
300	170	400	4800	–
450	350	1000	12000	–

Table 7: Compute allowance for the four road cases (seconds). FleetRoute (FL) values are hard algorithmic ceilings per seed, identical across Legacy, V3.1, and Current, with six configured software workers and `-no_improve_stop_s 0`. OR-Tools values are per-configuration limits and the corresponding 12-configuration allowance ceilings. Full observed sweep elapsed is retained only for the 100-node case; "–" means not retained, not zero.

Solver	Complete triples	Best/tied	Mean excess
Current	51	47	0.19%
V3.1	51	13	2.22%
Legacy	51	9	11.23%

Table 8: Strict parent-original cross-generation comparison. A "best/tied" count is computed within each complete instance/seed triple.

Comparator	Wins	Ties	Losses	Mean reduction
V3.1	38	9	4	1.90%
Legacy	42	9	0	8.41%

Table 9: Pairwise descriptive comparison on the same 51 strict parent-original instance/seed triples.

Appendix D: Application-Hardening Regression Detail

A fresh external Geofabrik country download was unavailable in the verification environment. Controlled HTTP regressions instead cover transfer, checksum, resume, corruption/fail-closed behavior, and metadata fallback; live external availability remains network-dependent.

Check	Recorded result
Release integrity	48/48 embedded files match byte count and SHA-256; decoded source snapshot passes.
Language syntax	Python compile-all PASS; JavaScript syntax PASS; C++17 syntax PASS for cluster and solver sources.
Road-access exactness	100 queries agree with exhaustive access to maximum distance error 1.16×10^{-10} m; 450-query batch 0.505 s.
Bulk import/access	450-row import 0.036 s; 451 access validations 0.426 s.
Graph materialization	450-customer fixture first materialization 0.250 s; cache reuse 0.017 s.
Interrupted PBF	Controlled HTTP regression resumes from byte 1,048,576 and verifies the final 5.4 MB payload.
Cold API/zero-to-route	Cold health/auth/state PASS; controlled 40-customer solve publishes route version 2 with four independently validated routes.

Table 10: Selected v8.1 hardening regressions. The results support packaged application reliability, not generalized production throughput.

Appendix E: Operational Workflow Evidence

The main paper uses four screens to establish the critical state chain from validated convergence through publication, driver-facing consumption, and field feedback. The remaining screenshots are grouped by function rather than repeated as near-duplicate evidence. Browser chrome and operating-system taskbars are cropped where appropriate; application state, route geometry, map content, and logs are otherwise unchanged.

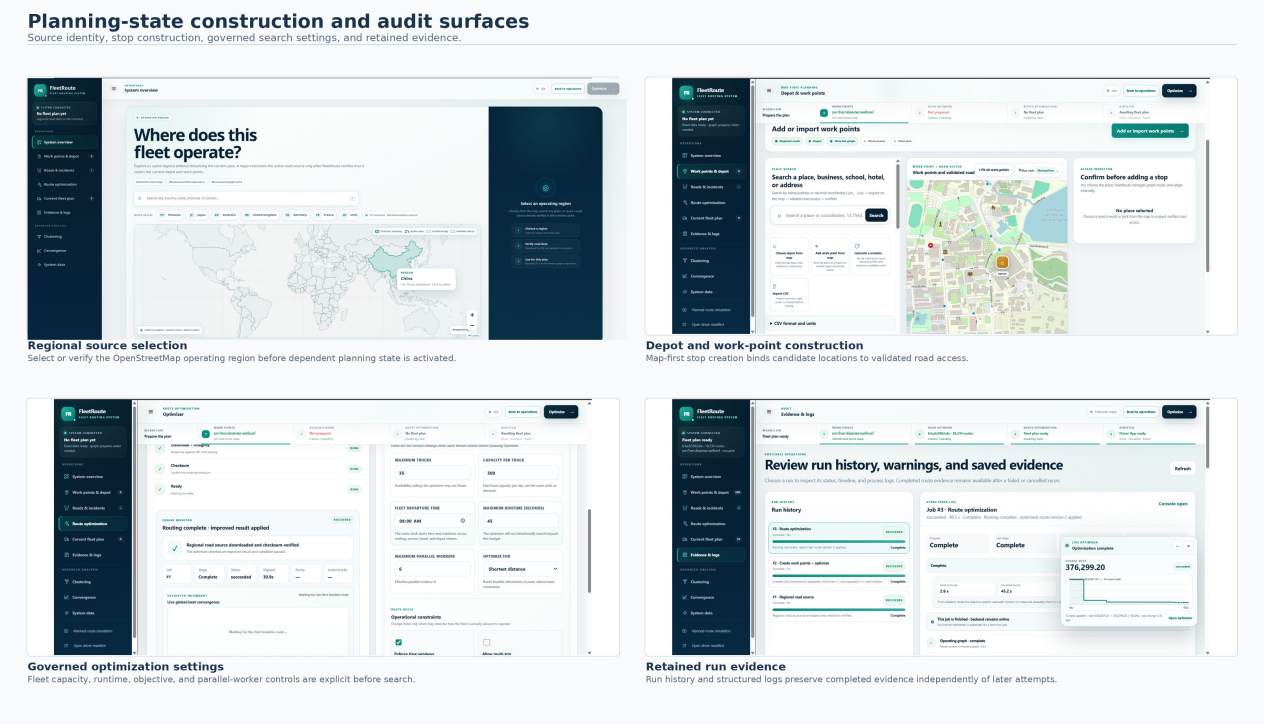


Figure 4: Planning-state construction and audit surfaces. Region selection establishes road-source identity; depot/work-point construction binds locations to road access; optimization settings expose the governed search contract; and run history retains completed evidence independently of later attempts.

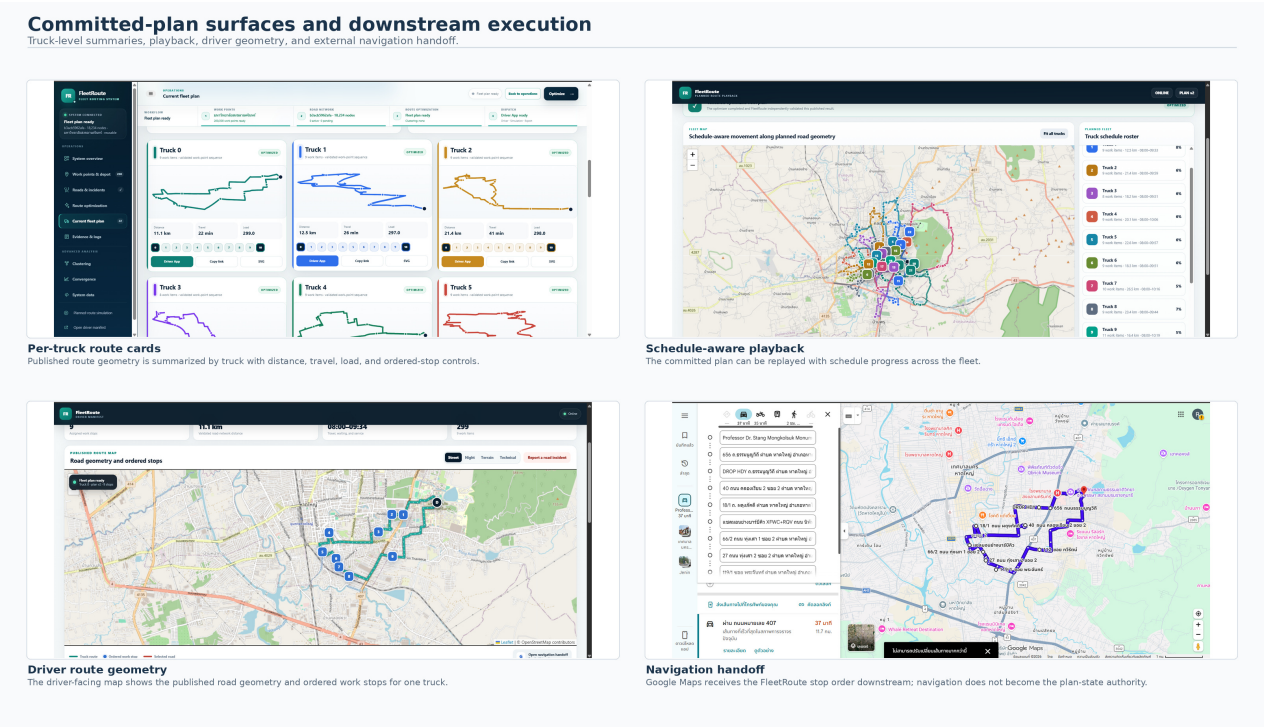


Figure 5: Committed-plan surfaces. Per-truck route cards, schedule-aware playback, the driver route map, and Google Maps navigation handoff all consume the committed FleetRoute revision. Navigation is downstream assistance, not an authority to reorder the audited stop sequence.

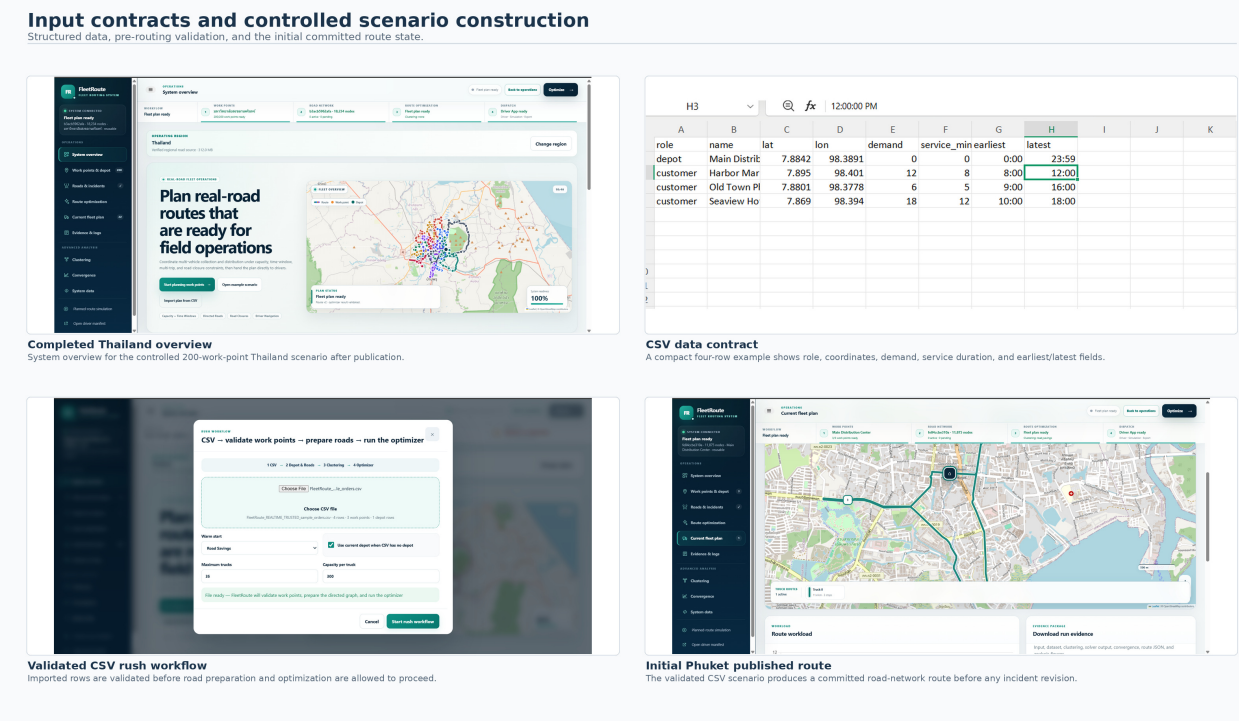


Figure 6: Input contracts and controlled scenario construction: the completed 200-work-point Thailand overview, compact Phuket CSV schema, validated rush import, and initial published Phuket route before any incident revision.

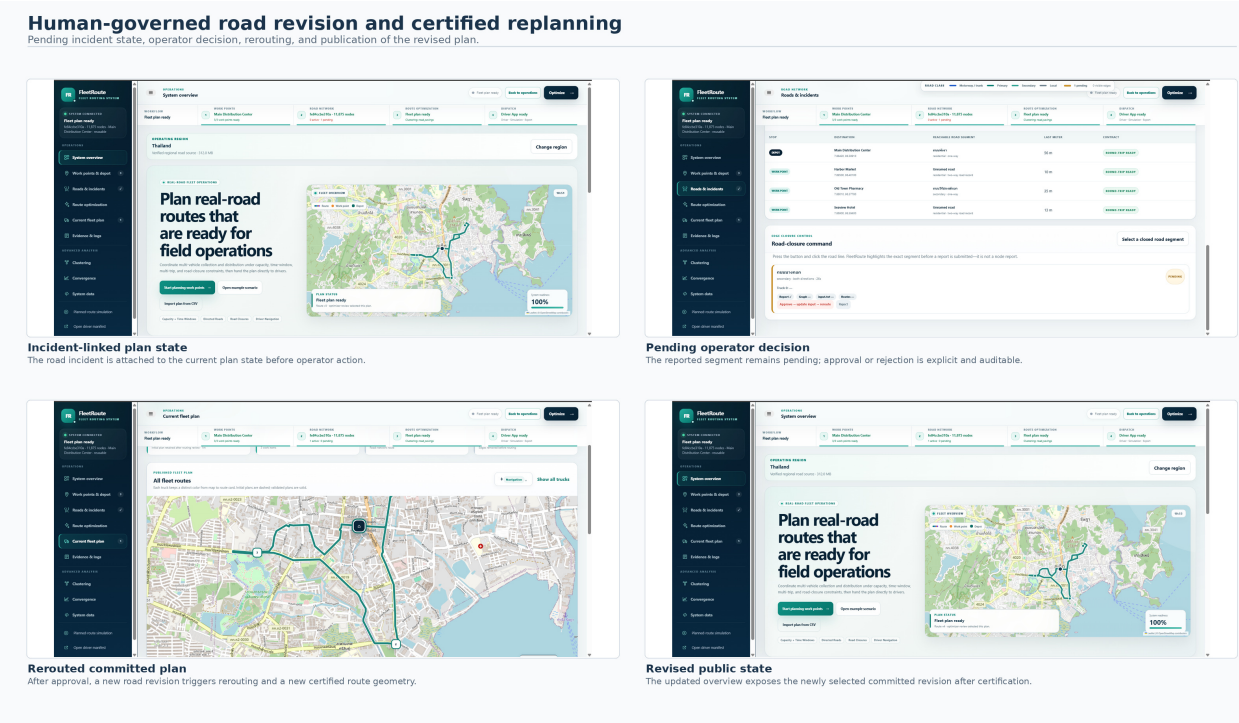


Figure 7: Human-governed replanning. An incident is attached to the current plan state and remains pending until operator action. Approval advances the road revision and triggers rerouting; the system exposes the replacement plan only after independent certification and commit.

Appendix F: Pilot Measurement and Acceptance Protocol

A field pilot should measure optimization value and operational reliability separately: a routing system can fail through stale state, weak change control, or user friction even when its route objective is strong. At minimum, the study should capture planner effort, route execution, publication integrity, incident response, driver behavior, and operational reliability. The aim is to compare FleetRoute with the organization's incumbent process rather than impose universal thresholds, while treating any loss of certification or change-control integrity as a critical failure. Successful and failed replans should both be reported.

Pilot dimension	Measurement, practical-value test, and failure signal
Planner effort	**Measure:** preparation time, manual edits, abandoned/restarted plans. **Value:** lower or comparable planning burden with fewer manual corrections. **Failure:** more operator work, repeated restarts, or workaround dependence.
Route execution	**Measure:** vehicles used; planned/realized distance and time; late/missed stops; operating cost. **Value:** improvement or non-inferiority on agreed service constraints. **Failure:** offline gains disappear in realized execution or service quality degrades.
Publication integrity	**Measure:** stale/invalid candidates blocked, incumbent retained, certification receipts. **Value:** zero uncertified publications and explainable plan transitions. **Failure:** any stale, invalid, or untraceable route becomes public.
Incident response	**Measure:** report-to-certified-plan latency, failed replans, incumbent-retention duration. **Value:** bounded recovery while a trusted incumbent remains available. **Failure:** road change bypasses operator approval or creates an unsafe service gap.
Driver behavior	**Measure:** route-order deviation, navigation errors, task completion, training/usability feedback. **Value:** low unplanned deviation and acceptable training burden. **Failure:** frequent reordering, navigation conflict, or workflow abandonment.
Reliability / governance	**Measure:** source/download failures, access-control events, audit retention, backup/rollback tests. **Value:** failures are detected, recoverable, and preserved in the audit trail. **Failure:** silent corruption, missing evidence, failed rollback, or unowned operational responsibility.

Table 11: Proposed pilot evidence contract. Operational value and decision-pipeline reliability are measured separately so route-quality gains cannot mask governance or usability failures.

Appendix G: Claim-to-Evidence Traceability

The supplement is structured so major numerical statements can be checked without reverse-engineering screenshots or solver logs. Raw parent and recovery archives remain authoritative; derived tables are rebuildable views. This appendix supports separate inspection of algorithmic competence, workflow validity, and software reliability. A minimal verification path is package manifest → notebook/source identity → frozen-parent benchmark identity and common-evaluator replay → rebuilt paper tables; the runnable Colab notebook is the reviewer-facing application entry point.

Claim / evidence unit	Primary artifact → independent check
213/213 supplement-file manifest	`MANIFEST_SHA256.json` → `tools/verify_package.py`
Application payload/source identity (48 embedded files)	submitted v8.1 notebook + `app/APP_PROVENANCE.json` → `tools/verify_app_notebook.py`
198 configured parent runs / 22 instances	`benchmark/raw/VRP_3SOLVER_STRICT_FAIR_V7_6THREAD_V31_TW_FIXED_FULL_RESULTS.zip` → `tools/verify_parent_benchmark.py`
180 originally valid parent routes	parent routes + `benchmark/evaluator/common_evaluator_reference.py` → `tools/recompute_all_common_objectives.py`
Paper feasibility / coverage / paired statistics	rebuilt feasibility, unresolved, and paired tables → `tools/rebuild_tables.py`
Public-reference and real-road comparisons	`tables/current_public_reference_gaps.csv`; `tables/osm_four_way_common_metric.csv` → rebuild + common-evaluator replay
Parent compute / stopping contract	`benchmark/compute_fairness/parent_experiment_manifest.json`; `benchmark/compute_fairness/fleetroute_budget_contract.csv` → `tools/verify_compute_fairness.py`
OR-Tools compute allowance	`benchmark/compute_fairness/ortools_osm_compute_allowance.csv`; retained 100-node raw output → `tools/verify_compute_fairness.py`
OR-Tools normalization provenance	`benchmark/ortools/OR_TOOLS_PROVENANCE.json` → replay on byte-identified OSM inputs
v8.1 hardening fixtures	`app/SYSTEM_HARDENING_AUDIT.json` → release audit receipts / integration tests

Table 12: Traceability from paper claims to primary artifacts and verification paths.

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